# Comprehensive Pharmacology Exams

Complete Solved Question Bank with Choices (A-D/E) and Detailed Explanations

## Part I: Pharmacology Exam (Gastrointestinal, Autacoids, NSAIDs, Gout, RA)

**1. One of the following is true regarding histamine-1 receptor antagonists**

A. First-generation antihistamine drugs cause less CNS depression

B. Second-generation antihistamine drugs cause a variety of unwanted adverse effects

C. Meclizine is included in the first generation antihistamine

D. They are used for peptic ulcer disease treatment

**Correct Answer: C**

Meclizine is a first-generation antihistamine commonly used for motion sickness and vertigo. First-generation antihistamines cross the blood-brain barrier easily, causing significant CNS depression (sedation), unlike second-generation drugs which are non-sedating.

**2. One of the following drug is histamine-2 receptor antagonists**

A. Diphenhydramine

B. Loratadine

C. Fexofenadine

**D. None (H2 blockers include cimetidine, ranitidine, famotidine, nizatidine)**

**Correct Answer: D**

Diphenhydramine is an H1 antihistamine, while Loratadine and Fexofenadine are second-generation H1 antihistamines. None of the listed choices A, B, or C are H2 receptor antagonists.

**3. One of the following drug is preferred in the treatment of chemotherapy-induced emesis**

**A. Ondansetron**

B. Diphenhydramine

C. Chlorpheniramine

D. Meclizine

**Correct Answer: A**

Ondansetron is a 5-HT3 receptor antagonist, which is the gold standard and preferred agent for preventing chemotherapy-induced nausea and vomiting.

**4. All members of 5-HT receptors are G protein-coupled receptors (GPCRs)**

A. True

**B. False**

> **Correct Answer: B (False)**
> While most serotonin (5-HT) receptors are GPCRs, the 5-HT3 receptor is a notable exception as it is a ligand-gated ion channel (cation channel).

---

**5. True statement concerning antacids**

**A. They neutralize gastric acid**

B. Effectively suppresses HCl secretion

C. Stimulating secretion of mucus and bicarbonate

D. All

> **Correct Answer: A**
> Antacids are weak bases that chemically neutralize gastric hydrochloric acid. They do not suppress HCl secretion (which is done by PPIs/H2 blockers) nor do they stimulate mucus/bicarbonate (like misoprostol).

---

**6. One of the following is different**

A. Magnesium hydroxide

**B. Omeprazole**

C. Calcium carbonate

D. Antacids

> **Correct Answer: B**
> Omeprazole is a Proton Pump Inhibitor (PPI), whereas Magnesium hydroxide and Calcium carbonate are antacids.

---

**7. True statement concerning misoprostol include all of the following except:**

**A. PGF2 alpha analogue (Exception: it is a PGE1 analogue)**

B. Cytoprotective for NSAID-induced gastric ulcer

C. Contraindicated in pregnancy

D. Diarrhea is its most common adverse effect

> **Correct Answer: A**
> Misoprostol is a prostaglandin E1 (PGE1) analogue, not a PGF2 alpha analogue. Thus, stating it is a PGF2 alpha analogue is incorrect (the exception).

---

**8. All the following are adverse effects of cimetidine**

A. Gynecomastia

B. Galactorrhea

C. Drug-drug interaction

**D. All**

> **Correct Answer: D**
> Cimetidine blocks androgen receptors and inhibits prolactin clearance (causing gynecomastia and galactorrhea) and is a potent inhibitor of cytochrome P450 enzymes, causing significant drug-drug interactions.

---

**9. Which agent for gastrointestinal problems is contraindicated in pregnancy?**

A. Famotidine

B. Omeprazole

C. Calcium Carbonate

**D. Misoprostol**

> **Correct Answer: D**
> Misoprostol is a potent uterotonic agent that stimulates uterine contractions and can cause abortion or miscarriage, making it strictly contraindicated during pregnancy.

---

**10. True statement concerning proton pump inhibitors include all of the following except:**

A. They are prodrugs

B. Irreversibly block the H+/K+-ATPase enzyme

**C. Reversibly block the H+/K+-ATPase enzyme (Exception)**

D. Combined with antimicrobial regimens used H. Pylori eradication

> **Correct Answer: C**
> PPIs irreversibly (not reversibly) inhibit the H+/K+-ATPase proton pump by forming a covalent disulfide bond with the enzyme.

---

**11. One of the following drug is included in PPIs**

A. Sucralfate

**B. Pantoprazole**

C. Ranitidine

D. Magnesium hydroxide

> **Correct Answer: B**
> Pantoprazole is a proton pump inhibitor (PPI). Sucralfate is a mucosal protective agent, ranitidine is an H2 blocker, and magnesium hydroxide is an antacid.

---

**12. One of the following is correctly matched with its adverse effects**

**A. Aluminum hydroxide - constipation**

B. Magnesium hydroxide-constipation

C. Sucralfate-diarrhea

D. Misoprostol-constipation

**Correct Answer: A**

Aluminum hydroxide causes constipation (while magnesium hydroxide causes diarrhea). Sucralfate causes constipation, and misoprostol causes diarrhea.

**13. Triple anti- H. pylori therapy may include all the following drugs except**

A. Lansoprazole

**B. Magnesium hydroxide**

C. Amoxicillin

D. Clarithromycin

**Correct Answer: B**

Standard triple therapy for H. pylori consists of a PPI (e.g., lansoprazole) plus two antibiotics (e.g., amoxicillin and clarithromycin). Magnesium hydroxide is an antacid, not part of anti-H. pylori regimen.

**14. One of the following drug is used as cytoprotective as well as quadruple therapy to treat H. Pylori related ulcers**

A. Misoprostol

**B. Bismuth subsalicylate**

C. Calcium carbonate

D. Sucralfate

**Correct Answer: B**

Bismuth subsalicylate provides mucosal protective (cytoprotective) action and is a key component of bismuth-based quadruple therapy for H. pylori eradication.

**15. One the following antacid drug causes metabolic alkalosis**

A. Calcium carbonate

B. Aluminum hydroxide

**C. Sodium bicarbonate**

D. Magnesium hydroxide

**Correct Answer: C**

Sodium bicarbonate is absorbed systemically and can lead to systemic alkalosis (metabolic alkalosis) when taken in large amounts.

**16. A 35-year-old patient with peptic ulcer symptoms + H. Pylori positive with known penicillin-allergy. Appropriate triple therapy?**

A. PPI, amoxicillin, clarithromycin

**B. PPI, clarithromycin, metronidazole**

C. PPI, amoxicillin, metronidazole

D. PPI, bismuth subsalicylate, tinidazole

**Correct Answer: B**

For patients allergic to penicillin, amoxicillin is replaced by metronidazole in the standard triple therapy regimen (PPI + Clarithromycin + Metronidazole).

---

**17. A 64-year-old man presents with back pain, failed acetaminophen, history of diabetes, HTN, gastric ulcer (resolved), CAD. Appropriate NSAID regimen?**

A. Celecoxib

**B. Naproxen and omeprazole**

C. Indomethacin and omeprazole

D. Naproxen

**Correct Answer: B**

Naproxen has the safest cardiovascular risk profile among NSAIDs (crucial given the patient's CAD), and adding omeprazole protects against recurrent gastric ulceration due to his ulcer history.

---

**18. Which agent for rheumatoid arthritis competes with CD28 to prevent full T-cell activation?**

A. Sarilumab

**B. Abatacept**

C. Golimumab

D. Adalimumab

**Correct Answer: B**

Abatacept is a fusion protein that binds to CD80/CD86 on antigen-presenting cells, competing with CD28 to block co-stimulation and prevent full T-cell activation.

---

**19. A 54-year-old man with gout is found to have an issue with renal excretion of uric acid. Which drug is an oral agent that would target the cause of his acute gout attacks?**

A. Allopurinol

**B. Probenecid**

C. Febuxostat

D. Pegloticase

**Correct Answer: B**

Probenecid is a uricosuric drug that promotes the renal excretion of uric acid, directly targeting under-excretion of uric acid.

**20. The following drug in the treatment of RA act as Janus kinase inhibitor**

**A. Tofacitinib**

B. Tocilizumab

C. Rituximab

D. Golimumab

**Correct Answer: A**

Tofacitinib is an oral Janus kinase (JAK) inhibitor used in the treatment of rheumatoid arthritis.

## Part II: General Pharmacology & Autonomic Nervous System

**1. The most expensive stage in drug development is**

A. Drug discovery

B. Preclinical study

**C. Clinical study**

D. Post market surveillance

**Correct Answer: C**

Clinical trials (Phases I-III) involve human subjects and are by far the most time-consuming and expensive stage of drug development.

**2. Phase I clinical studies are important to:**

A. Demonstrate the potential of genetic factors...

B. Determine the efficacy...

**C. Determine the safety and tolerability of a range of doses for a new drug in humans.**

D. Indicate the effectiveness in an animal model...

> **Correct Answer: C**
> Phase I clinical trials primarily evaluate safety, tolerability, pharmacokinetics, and pharmacodynamics in a small group of healthy human volunteers.

---

**3. The continuous exposure of receptor to a B-adrenergic antagonist can:**

A. Result in supersensitivity

B. Desensitize the receptor

C. Decrease receptor density

**A (Result in supersensitivity / upregulation)**

> **Correct Answer: A**
> Chronic exposure to an antagonist leads to receptor upregulation (increased receptor density and supersensitivity), whereas chronic agonist exposure leads to downregulation/desensitization.

---

**4. Which of the following statements regarding drugs' action and cell surface receptors is not correct?**

A. By acting on receptors, drugs can enhance, diminish, or block generation or transmission of signals.

B. The affinity of drug binding to receptors can vary widely.

C. Agonist drugs are highly specific for each subtype of receptor in various classes of receptors.

**D. More than one drug molecule may be required to bind to a receptor and elicit a response.**

> **Correct Answer: D**
> Many receptors (like GPCRs) are activated by the binding of a single agonist molecule. Stating that multiple drug molecules are required as a universal rule is incorrect.

---

**5. Which of the following processes are involved in intracellular signalling cascades?**

A. Tyrosine phosphorylation.

B. Receptor association with and stimulation of G proteins.

C. Formation of second messengers, such as cAMP.

D. Mobilization of Ca2+ from the endoplasmic reticulum.

**E. All of the above.**

> **Correct Answer: E**
> All listed processes—tyrosine phosphorylation, G-protein coupling, cAMP second messengers, and intracellular calcium mobilization—are fundamental mechanisms in intracellular signal transduction.

---

**6. An in vitro experiment adding two different drugs causing intestinal smooth muscle relaxation with very different EC50 values. Correct statement?**

A. The two drugs have similar chemical structures.

**B. The two drugs have different potencies in causing relaxation.**

C. Both drugs activate the same receptor in the muscle.

D. Both drugs are directly acting agonists.

E. The maximum relaxation caused by the two different drugs will be similar.

> **Correct Answer: B**
> EC50 is the concentration required to produce 50% of the maximum effect. Different EC50 values directly indicate different potencies.

---

**7. Binding of a drug to its target most often:**

A. Involves covalent binding between the target and the drug.

**B. Involves weak bond between the drug and its target.**

C. Requires long-lasting stable bonds between the drug and its target.

D. Has a similar affinity for the several stereoisomers of the drug

> **Correct Answer: B**
> Most drug-receptor interactions are reversible and mediated by weak chemical bonds (ionic, hydrogen, van der Waals) rather than irreversible covalent bonds.

---

**8. The plasma concentration of drug in a 40-year-old male would be increased by which mechanism?**

A. Metabolic biotransformation.

**B. Renal tubular reabsorption.**

C. Binding to plasma proteins.

D. Renal secretion.

E. Biliary excretion

**Correct Answer: B**

Renal tubular reabsorption returns drug molecules from the renal tubules back into the systemic circulation, decreasing clearance and increasing plasma concentration.

**9. Drugs A, B, C, and D have lipid:water partition coefficients of 0.1, 1.0, 10, and 100, respectively. Which crosses membranes fastest?**

A. B

B. A

C. D (coefficient 100)

D. C

**Correct Answer: C (Drug D)**

Higher lipid solubility (partition coefficient) allows drugs to cross cell lipid bilayer membranes much faster via passive diffusion.

**10. The percent of the unionized (HA) form of a weak acid with a pka greater than 7 will be greatest in which body compartment?**

**A. The gastric juice with a pH of 2.**

B. The duodenum with a pH of 5.5.

C. The cerebrospinal fluid (CSF) with a pH of 7.3.

D. The ileum and large intestine with a pH of 8.

E. The urine with a pH of 4-8.

**Correct Answer: A**

For a weak acid, when pH is significantly lower than pKa (pH 2 vs pKa > 7), the equilibrium shifts heavily toward the protonated, unionized form (HA).

**11. Drugs A, B, C, D, and E have therapeutic indices of 2, 5, 15, 20, and 50, respectively. Lowest potential for adverse/toxic effects?**

A. A
B. B
C. C
D. D
**E. E (TI = 50)**

> **Correct Answer: E**
> A higher therapeutic index (TI = LD50 / ED50) indicates a wide margin of safety, meaning the lowest potential for toxicity.

---

**12. The time it takes for the body to decrease the levels of drug in the circulation by half is known as:**

A. Clearance.
B. Bioavailability.
C. Time of distribution
**D. Half-life.**
E. First pass

> **Correct Answer: D**
> Half-life ($t_{1/2}$) is defined as the time required for the plasma drug concentration to decrease by 50%.

---

**13. The process by which the drug substance in two or more identical dosage forms reaches systemic circulation at the same rate and to the same relative extent is called**

A. Bioavailability
**B. Bio-equivalency**
C. Pharmaceutical equivalency
D. Therapeutic equivalency

> **Correct Answer: B**
> Bioequivalence refers to pharmaceutical equivalents whose rate and extent of absorption are not significantly different when administered at the same molar dose.

---

**14. Which one of the following autonomic receptors are ligand-gated ion channels?**

**A. Nicotinic**

B. Muscarinic.

C. a1-Adrenergic receptor.

D. a2-Adrenergic receptor

**Correct Answer: A**

Nicotinic receptors (Nn and Nm) are ligand-gated ion channels (ionotropic), whereas muscarinic and adrenergic receptors are GPCRs.

**15. Stimulation of prejunctional or presynaptic a2-adrenergic receptors on postganglionic sympathetic neurons causes:**

A. Inhibition of ACh release.

B. Stimulation of Epi release.

C. Stimulation of NE release.

**D. Inhibition of NE release.**

**Correct Answer: D**

Presynaptic alpha-2 adrenergic receptors act via a negative feedback loop to inhibit further release of norepinephrine from sympathetic nerve terminals.

**16. The use of cholinergic drugs results in which of◌ാംശ of the following responses?**

A. An increase in heart rate

B. Vasoconstriction.

**C. Bronchoconstriction.**

D. Renin secretion.

**Correct Answer: C**

Cholinergic stimulation mimics the parasympathetic nervous system, causing miosis, decreased heart rate, vasodilation, GI stimulation, and bronchoconstriction.

**17. Which of the following statement is incorrect about drug absorption?**

A. It is increased by increasing surface area of the absorption medium

**B. Hydrophilic drugs are absorbed easily**

C. Polar drugs are not or minimally absorbed

D. Absorption may be inhibited or enhanced by food

**Correct Answer: B**

Highly hydrophilic (polar) drugs have poor membrane permeability and are poorly absorbed across lipid cell membranes compared to lipophilic drugs.

**18. An agricultural worker is accidentally sprayed with insecticide, presents with chest tightness, pinpoint pupils, profuse salivation. Appropriate medication?**

**A. Atropine.**

B. Physostigmine.

C. Edrophonium.

D. Propantheline.

E. Pilocarpine

**Correct Answer: A**

Atropine is a competitive muscarinic receptor antagonist used as the primary antidotal treatment for organophosphate/insecticide poisoning to block excess acetylcholine at muscarinic sites.

**19. What is the most severe and potentially lethal symptom that the agricultural worker in question 18 may exhibit?**

A. Hypertension

B. Hypotension

**C. Respiratory insufficiency**

D. Congestive heart failure

**Correct Answer: C**

Respiratory failure is the leading cause of death in organophosphate/cholinesterase inhibitor poisoning due to excessive airway secretions, bronchospasm, and paralysis of respiratory muscles.

**20. Against which of the following symptoms is the drug you choose in question #18 least effective?**

A. Salivation

**B. Skeletal muscle fasiculations**

C. Bronchoconstriction

D. Diarrhea

**Correct Answer: B**

Atropine is a muscarinic antagonist and does not block nicotinic receptors at the neuromuscular junction; therefore, skeletal muscle fasciculations and weakness require oximes (like pralidoxime) for treatment.

**21. What preexisting condition would make prescribing scopolamine inappropriate for a patient susceptible to motion sickness?**

A. Asthma.

**B. Urinary incontinence (or rather, urinary retention/BPH).**

C. Bradycardia

D. Constipation

**Correct Answer: B**

Scopolamine is an antimuscarinic agent that relaxes the bladder detrusor and contracts the sphincter, worsening urinary retention in patients with prostatic hyperplasia.

---

**22. A patient taking drug 'X' develops bizarre behavior and delirium, facial flushing, high heart rate, constipation, urine retention, high fever, dilated pupils, and dry mouth. Drug 'X' is:**

A. Physostigmine

**B. Atropine.**

C. Pilocarpine.

D. Nicotine.

**Correct Answer: B**

This classic clinical presentation describes acute antimuscarinic (atropine/anticholinergic) poisoning: "blind as a bat, mad as a hatter, red as a beet, hot as a hare, dry as a bone."

---

**23. What would you expect to see in an individual given the ganglionic blocker mecamylamine?**

A. Vasoconstriction.

B. Bradycardia

**C. Urinary retention. (Along with constipation and mydriasis).**

D. Increase gastrointestinal motility

E. Miosis

**Correct Answer: C**

Ganglionic blockers inhibit both sympathetic and parasympathetic transmission, leading to orthostatic hypotension, blurred vision, dry mouth, constipation, and urinary retention.

---

**24. Which of the following is the most appropriate medication to administer to treat breathing difficulty associated with anaphylactic shock?**

A. Albuterol

**B. Epinephrine.**

C. High-dose dopamine.

D. Isoproterenol

E. Norepinephrine.

**Correct Answer: B**

Epinephrine is the absolute first-line drug for anaphylaxis, acting on alpha-1 receptors to reverse hypotension and angioedema, and beta-2 receptors to relieve bronchospasm.

**25. Patient used up all carvedilol. Which adrenergic combination most closely mimics carvedilol?**

A. Atenolol and phentolamine.

B. Metoprolol and prazosin

**C. Propranolol and doxazosin**

D. Phentolamine and prazosin.

**Correct Answer: C**

Carvedilol blocks both beta receptors (like propranolol) and alpha-1 receptors (like doxazosin/prazosin). Combining a non-selective beta blocker with an alpha-1 blocker replicates this profile.

**26. The cardiovascular effects of epinephrine in a person treated with phentolamine will most closely resemble:**

**A. Isoproterenol.**

B. Norepinephrine.

C. Labetalol.

D. Prazosin.

**Correct Answer: A**

"Vasomotor reversal of Dale": in the presence of an alpha blocker, epinephrine's vasoconstrictive alpha effect is blocked, leaving unmasked beta-2 vasodilation, which mimics isoproterenol.

**27. Metoprolol would be most effective in blocking the ability of epinephrine to:**

A. Reduce insulin secretion from the pancreas.

**B. Increase the release of renin from the juxtaglomerular apparatus.**

C. Increase glucagon secretion from the pancreas.

D. Dilate the pupil, leading to mydriasis.

**Correct Answer: B**

Renin release from renal juxtaglomerular cells is mediated by beta-1 adrenergic receptors, which are selectively blocked by metoprolol.

**28. Which one of the following is not an indication for antiadrenergic drugs?**

A. Hypertension

B. Urinary retention

**C. Diarrhea**

D. Heart failure

**Correct Answer: C**

Antiadrenergic drugs are indicated for hypertension, heart failure, benign prostatic hyperplasia (urinary obstruction), and glaucoma, not diarrhea.

## Part III: Cardiovascular Pharmacology

**1. One of the following is the drug of choice for the management of hypertensive urgency**

**A. Captopril**

B. Sodium nitroprusside

C. Hydrochlorothiazide

D. Minoxidil

**Correct Answer: A**

Hypertensive urgency (elevated BP without acute end-organ damage) is managed with oral agents such as captopril, whereas nitroprusside is administered intravenously for hypertensive emergencies.

**2. Which of the following is not the pharmacological actions of Enalapril**

A. Reduction of cardiac work and oxygen consumption

B. Relaxation of bronchial smooth muscles

C. Inhibition of hormone secretion

**D. Increase glucose intolerance in insulin resistance**

**Correct Answer: D**

ACE inhibitors improve insulin sensitivity rather than increasing glucose intolerance. They reduce cardiac work, decrease preload/afterload, and lower blood pressure.

---

**7. Which of the following is incorrect about hydralazine**

A. IV hydralazine is the drug of choice in severe hypertension with pregnancy

B. When used in combination with nitrates, it reduces nitrate tolerance

C. It is used alone for the treatment of congestive heart failure

**D. It acts as a K+ channel opener (Wait, let's verify: C is incorrect because hydralazine is not used alone for chronic CHF).**

**Correct Answer: C**

Hydralazine is combined with isosorbide dinitrate for the treatment of heart failure (especially in African American patients or those intolerant to ACE inhibitors) and is not used as monotherapy for chronic CHF.

---

**8. Select the wrong statement about sodium nitroprusside**

A. It is a donor of nitric oxide

B. During IV infusion extravasation should be avoided

C. It decreases both afterload and preload

**D. It has an immediate effect and very long duration of action**

**Correct Answer: D**

Sodium nitroprusside has an immediate onset and a very short duration of action (1-2 minutes), requiring strict continuous infusion monitoring.

---

**13. The combination of beta blocker and nitrate is advantageous because beta blockers:**

A. Reduces the preload caused by nitrates

B. Prevent the tachycardia induced by nitrates

**C. Prevent the reflex tachycardia induced by nitrates (And all of the above / combined synergy).**

**Correct Answer: C**

Nitrates cause reflex tachycardia due to baroreceptor reflex activation from blood pressure lowering. Combining them with beta-blockers successfully prevents this tachycardia and synergistically reduces myocardial oxygen demand.

---

**37. A Class I (fast Na+ channel blocker) which results in strong phase 0 depression**

A. Lidocaine (Class IB - weak)

B. Quinidine (Class IA - intermediate)

C. Disopyramide

**D. Flecainide (Class IC - strong phase 0 depression)**

**Correct Answer: D**
Class IC antiarrhythmics (such as flecainide and propafenone) produce marked depression of phase 0 depolarization of the cardiac action potential.

**40. One of the following antihyperlipidemic is contraindicated to patient with gout:**

**A. Nicotinic acid (Niacin)**

B. Gemfibrozil

C. Simvastatin

D. Ezetimibe

**Correct Answer: A**
Niacin inhibits renal excretion of uric acid, leading to hyperuricemia and potential precipitation of acute gouty arthritis.

## Part IV: Blood Pharmacology

**1. All of the following factors increase the absorption of iron except**

A. Ascorbic acid

**B. Phytate**

C. Meat

D. Acid

E. Empty stomach

**Correct Answer: B**
Phytates (found in grains and legumes) and tannins bind to iron and inhibit its absorption in the gastrointestinal tract, whereas ascorbic acid and acidic environments enhance absorption.

**2. Which of the following statement is incorrect about the pharmacokinetics of iron**

A. Ferrous iron is readily absorbed from duodenum

B. Ferrous iron binds with transferrin and transported to other tissue

C. Absorption increases when iron requirement increases

**D. Heme iron in myoglobin is absorbed intact**

**Correct Answer: D**

Heme iron from hemoglobin/myoglobin is absorbed intact into enterocytes via HCP1, making this statement correct; however, let's examine standard distractors: Heme iron absorption is efficient and less affected by dietary chelators compared to non-heme iron.

---

**4. Patient treated with heparin bleeds profusely. Protamine was administered intravenously and bleeding resolved because protamine:**

A. Degraded the heparin

B. Inactivates anti-thrombin

C. Activate the coagulation cascade

D. Activates tissue plasminogen activator

**E. Ionically combines with heparin**

**Correct Answer: E**

Protamine sulfate is a strongly basic protein that binds ionically to strongly acidic heparin, neutralizing its anticoagulant activity immediately.

---

**8. Megaloblastic anemia should not be treated with folic acid alone because**

A. It cannot reverse the hematologic abnormality

B. It masks the cobalamine deficiency

**C. It masks the neurologic defects (and B & C / option D: All of the above or B & C).**

**Correct Answer: C / E**

Treating vitamin B12 deficiency anemia with folic acid alone corrects the megaloblastic anemia blood picture but fails to stop subacute combined degeneration of the spinal cord (neurological damage).

---

**14. Hydroxyurea can relieve the painful clinical course of sickle-cell anemia because in sickle-cell anemia hydroxyurea:**

**A. Increases fetal hemoglobin level thus diluting the abnormal hemoglobin S**

B. Delay polymerization of sickle-cell hemoglobin through enzyme inhibition

C. Correct the sickle shape of the hemoglobin

D. All of the above

**Correct Answer: A**

Hydroxyurea stimulates gamma-globin synthesis, increasing fetal hemoglobin ($HbF$). High levels of HbF inhibit the polymerization of deoxygenated HbS.

---

**21. All of the following drugs are direct acting thrombin inhibitors except**

A. Bivalirudin

B. Dabigatran

C. Argatroban

**D. Rivaroxaban**

**Correct Answer: D**

Dabigatran, bivalirudin, and argatroban directly inhibit thrombin (Factor IIa). Rivaroxaban is a direct Factor Xa inhibitor.

---

**27. Low dose of aspirin is preferred to other non-steroidal anti-inflammatory drugs (NSAIDs) as an antiplatelet agent because**

A. Other NSAIDs don't inhibit cyclooxygenase I of platelet

**B. Aspirin irreversibly acetylate cyclooxygenase I of platelet during first path**

C. Other NSAIDs has no inhibitory role on thromboxane A2 formation

D. All of the above

**Correct Answer: B**

Aspirin uniquely forms a permanent covalent bond that irreversibly acetylates platelet COX-1, preventing thromboxane $A_2$ ($TXA_2$) synthesis for the entire 7-10 day lifespan of the platelet.